

YANG MA

CP3, UCLouvain $\diamond$ Chemin du Cyclotron 2, 1348 Louvain-la-Neuve, Belgium

✉ yang.ma@uclouvain.be 🌐 <https://yangphy.github.io>

PROFESSIONAL PREPARATION

Ph.D. in Physics

University of Pittsburgh, PA, U.S.A.

August 2022

Advisor: Tao Han

M.S. in Physics

Chongqing University, Chongqing, China

June 2016

SKILLS

Programming Language: C/C++, Fortran, Python, Shell script

Handy Programs: Mathematica, Matlab, L^AT_EX, Excel, Powerpoint, Linux (OS)

HEP Packages: MADGRAPH5_AMC@NLO, Pythia, FeynRules, FeynArts, FeynCalc, FormCalc, WHIZARD, ManeParse

POSITION HELD

F.R.S.-FNRS Postdoctoral Fellow (Chargé de recherches)

October 2024 - present

Center for Cosmology, Particle Physics and Phenomenology (CP3)

Université catholique de Louvain (UCLouvain), Belgium

Individual research fellowship awarded by the Belgian National Fund for Scientific Research (F.R.S.-FNRS).

International Postdoctoral Fellow

September 2022 - September 2024

INFN Sezione di Bologna, Italy

Awarded through a nationwide competition (1 of only 16 positions across all INFN divisions in Italy).

Graduate Teaching Fellow

May 2022 - July 2022

Dept. Physics & Astronomy, University of Pittsburgh

Graduate Research Fellow

January 2022 - April 2022

Dept. Physics & Astronomy, University of Pittsburgh

Arts & Sciences Pre-Doctoral Fellow

September 2020 - December 2021

Kenneth P. Dietrich School of Arts & Sciences, University of Pittsburgh

Graduate Research Assistant

January 2020 - August 2020

Dept. Physics & Astronomy, University of Pittsburgh

Graduate Teaching Assistant

August 2016 - January 2020

Dept. Physics & Astronomy, University of Pittsburgh

AWARDS AND HONORS

Outstanding Reviewer Awards 2022

March 2023

Journal of Physics Communications, IOP Publishing

DPF Student Travel Award

April 2022

APS Division of Particles and Fields (DPF)

Thomas-Lain Scholarship

April 2021

Dept. Physics & Astronomy, University of Pittsburgh

FGSA Award for Excellence in Graduate Research

February 2021

American Physical Society (APS)

Arts & Sciences Graduate Fellowship

September 2020

Kenneth P. Dietrich School of Arts & Sciences, University of Pittsburgh

Pitt Physics and Astronomy China Initiative (PACI) Scholarship

September 2016

Dept. Physics & Astronomy, University of Pittsburgh

MEMBERSHIP IN PROFESSIONAL SOCIETIES

American Physical Society (APS) member

March 2021 - present

International Organization of Chinese Physics & Astrophysics member

September 2021 - present

American Association for the Advancement of Science (AAAS) member

September 2021 - present

International Muon collider collaboration (IMCC) member

July 2022 - present

REFEREE SERVICE

- Chinese Physics C (*CPC*) $\times 3$
- Chinese Physics Letters (*Chin. Phys. Lett.*) $\times 1$
- The European Physical Journal C (*Eur. Phys. J. C*) $\times 1$
- Machine Learning: Science and Technology (*MLST*) $\times 1$
- Nuclear Physics B (*Nucl. Phys. B*) $\times 1$
- Journal of High Energy Physics (*JHEP*) $\times 1$
- Journal of Physics Communications (*J. Phys. Commun.*) $\times 1$
- Journal of Physics G: Nuclear and Particle Physics (*J. Phys. G*) $\times 2$
- Physics Letters B (*Phys. Lett. B*) $\times 1$
- Physical Review D (*Phys. Rev. D*) $\times 1$
- Physical Review Letters (*Phys. Rev. Lett.*) $\times 2$

TEACHING EXPERIENCE

Graduate Teaching Fellow (instructor) at the University of Pittsburgh

- PHYS 0174 - *Basic Physics, Science and Engineering 1*, Summer 2022
Covers Mechanics and Wave

Graduate Teaching Assistant at the University of Pittsburgh

- PHYS 0219 - *Basic Laboratory Physics for Science and Engineering*
Fall 2016, Spring 2017, and Fall 2018
- PHYS 0212 - *Introduction to Laboratory Physics*
Fall 2017, Spring 2018, and Summer 2018

- PHYS 0110 - *Introduction to Physics 1*, Summer 2018
Covers Mechanics, Heat and Thermodynamics, and Waves
- PHYS 0111 - *Introduction to Physics 2*, Summer 2017
Covers Thermodynamics, Electromagnetism, Optics, Special Relativity, and Quantum Physics
- PHYS 0175 - *Basic Physics, Science and Engineering 2*, Spring 2019 and Summer 2021
Covers Electromagnetism, Elementary Quantum Mechanics, and Atomic Structure

Graduate Teaching Assistant at Chongqing University

- College Physics I - *Classical Mechanics and Electromagnetism*, Spring 2014
- College Physics II - *Thermodynamics, Optics and Special Relativity*, Fall 2013

MENTORING EXPERIENCE

- Dept. Physics & Astronomy Graduate Student Mentor August 2021 - May 2022
Mentor three first year graduate students
- Assist to guide one visiting graduate student (*Xiaozhe Tan*) December 2019 - December 2020
(*JHEP* **08** (2022) 073 [2202.08273])
- Assist to guide one visiting undergraduate student June 2019 - August 2019

PUBLICATIONS IN REFEREED JOURNALS

1. D. Buttazzo, G. Levati, Y. Ma, F. Maltoni, P. Paradisi and Z. Wang, *Probing τ lepton dipole moments at future Lepton Colliders*, [2604.14281](#).
2. P. Brecht, T. Banno, M. Höfer, S. Iguro, W. Kilian, Y. Ma et al., *Chasing the Two-Higgs-Doublet Model via Electroweak Corrections at e^+e^- Colliders*, *Phys. Rev. Lett.* **136** (2026) 081801 [2509.05421].
3. F. Kling, Y. Ma, K. Mękała, J. Reuter and Z. Tabrizi, *Non-standard neutrino interactions at a muon collider neutrino detector*, *JHEP* **05** (2026) 050 [2508.00761].
4. S.-s. Bao, Y. Ma, Y. Wu, K. Xie and H. Zhang, *Light Axion-Like Particles at Future Lepton Colliders*, *JHEP* **10** (2025) 122 [2505.10023].
5. Y. Ma, D. Pagani and M. Zaro, *EW corrections and heavy boson radiation at a high-energy muon collider*, *Phys. Rev. D* **111** (2025) 053002 [2409.09129].
6. Y. Ma and Z. Wang, *New probe of dark matter-baryon interactions in compact stellar systems*, *Phys. Rev. D* **111** (2025) L061302 [2408.01818].
7. E. Celada, T. Han, W. Kilian, N. Kreher, Y. Ma, F. Maltoni et al., *Probing Higgs-muon interactions at a multi-TeV muon collider*, *JHEP* **08** (2024) 021 [2312.13082].
8. C. Accettura et al., *Towards a muon collider*, *Eur. Phys. J. C* **83** (2023) 864 [2303.08533].
9. T. Han, A. K. Leibovich, Y. Ma and X.-Z. Tan, *Higgs boson decay to charmonia via c -quark fragmentation*, *JHEP* **08** (2022) 073 [2202.08273].
10. T. Han, W. Kilian, N. Kreher, Y. Ma, J. Reuter, T. Striegl et al., *Precision Test of the Muon-Higgs Coupling at a High-energy Muon Collider*, *JHEP* **12** (2021) 162 [2108.05362].
11. D. Buarque et al., *Vector Boson Scattering Processes: Status and Prospects*, *Rev. Phys.* **8** (2022) 100071 [2106.01393].

12. T. Han, Y. Ma and K. Xie, *Quark and gluon contents of a lepton at high energies*, *JHEP* **02** (2022) 154 [[2103.09844](#)].
13. T. Han, Y. Ma and K. Xie, *High energy leptonic collisions and electroweak parton distribution functions*, *Phys. Rev. D* **103** (2021) L031301 [[2007.14300](#)].
14. Z. Sun and Y. Ma, *Inclusive productions of $\Upsilon(1S, 2S, 3S)$ and $\chi_b(1P, 2P, 3P)$ via the Higgs boson decay*, *Phys. Rev. D* **100** (2019) 094019 [[1909.08548](#)].
15. Z. Sun, X.-G. Wu, Y. Ma and S. J. Brodsky, *Exclusive production of $J/\psi + \eta_c$ at the B factories Belle and Babar using the principle of maximum conformality*, *Phys. Rev. D* **98** (2018) 094001 [[1807.04503](#)].
16. Y. Ma and X.-G. Wu, *Renormalization scheme dependence of high-order perturbative QCD predictions*, *Phys. Rev. D* **97** (2018) 036024 [[1707.09886](#)].
17. J.-M. Shen, X.-G. Wu, Y. Ma and S. J. Brodsky, *The Generalized Scheme-Independent Crewther Relation in QCD*, *Phys. Lett. B* **770** (2017) 494 [[1611.07249](#)].
18. H.-Y. Bi, X.-G. Wu, Y. Ma, H.-H. Ma, S. J. Brodsky and M. Mojaza, *Degeneracy Relations in QCD and the Equivalence of Two Systematic All-Orders Methods for Setting the Renormalization Scale*, *Phys. Lett. B* **748** (2015) 13 [[1505.04958](#)].
19. H.-H. Ma, X.-G. Wu, Y. Ma, S. J. Brodsky and M. Mojaza, *Setting the renormalization scale in perturbative QCD: Comparisons of the principle of maximum conformality with the sequential extended Brodsky-Lepage-Mackenzie approach*, *Phys. Rev. D* **91** (2015) 094028 [[1504.01260](#)].
20. Y. Ma, X.-G. Wu, H.-H. Ma and H.-Y. Han, *General Properties on Applying the Principle of Minimum Sensitivity to High-order Perturbative QCD Predictions*, *Phys. Rev. D* **91** (2015) 034006 [[1412.8514](#)].
21. H.-B. Fu, X.-G. Wu and Y. Ma, *$B \rightarrow K^*$ Transition Form Factors and the Semi-leptonic Decay $B \rightarrow K^* \mu^+ \mu^-$* , *J. Phys. G* **43** (2016) 015002 [[1411.6423](#)].
22. H.-B. Fu, X.-G. Wu, H.-Y. Han, Y. Ma and H.-Y. Bi, *The ρ -meson longitudinal leading-twist distribution amplitude*, *Phys. Lett. B* **738** (2014) 228 [[1409.3053](#)].
23. G. Chen, X.-G. Wu, Z. Sun, Y. Ma and H.-B. Fu, *Photoproduction of doubly heavy baryon at the ILC*, *JHEP* **12** (2014) 018 [[1408.4615](#)].
24. H.-B. Fu, X.-G. Wu, H.-Y. Han and Y. Ma, *$B \rightarrow \rho$ transition form factors and the ρ -meson transverse leading-twist distribution amplitude*, *J. Phys. G* **42** (2015) 055002 [[1406.3892](#)].
25. X.-G. Wu, Y. Ma, S.-Q. Wang, H.-B. Fu, H.-H. Ma, S. J. Brodsky et al., *Renormalization Group Invariance and Optimal QCD Renormalization Scale-Setting*, *Rept. Prog. Phys.* **78** (2015) 126201 [[1405.3196](#)].
26. S.-Q. Wang, X.-G. Wu, J.-M. Shen, H.-Y. Han and Y. Ma, *QCD improved electroweak parameter ρ* , *Phys. Rev. D* **89** (2014) 116001 [[1402.0975](#)].
27. Z. Sun, X.-G. Wu, G. Chen, Y. Ma, H.-H. Ma and H.-Y. Bi, *Bottomonium production associated with a photon at a high luminosity e^+e^- collider with the one-loop QCD correction*, *Phys. Rev. D* **89** (2014) 074035 [[1401.2735](#)].
28. H.-B. Fu, X.-G. Wu, H.-Y. Han, Y. Ma and T. Zhong, *$|V_{cb}|$ from the semileptonic decay $B \rightarrow D \ell \bar{\nu}_\ell$ and the properties of the D meson distribution amplitude*, *Nucl. Phys. B* **884** (2014) 172 [[1309.5723](#)].

OTHER PUBLICATIONS

1. MuCoL collaboration, *MuCol Milestone Report No. 7: Consolidated Parameters*, [2510.27437](#).
2. CEPC STUDY GROUP collaboration, *CEPC Technical Design Report - Reference Detector*, [2510.05260](#).
3. J. Altmann et al., *ECFA Higgs, electroweak, and top Factory Study*, vol. 5/2025 of *CERN Yellow Reports: Monographs*. 6, 2025, [10.23731/CYRM-2025-005](#), [[2506.15390](#)].
4. INTERNATIONAL MUON COLLIDER collaboration, *The Muon Collider*, [2504.21417](#).
5. MuCoL collaboration, *MuCol Milestone Report No. 5: Preliminary Parameters*, [2411.02966](#).
6. Y. Ma, E. Celada, T. Han, W. Kilian, N. Kreher, F. Maltoni et al., *Higgs-muon interactions at a multi-TeV muon collider*, *PoS ICHEP2024* (2024) 092 [[2410.06991](#)].
7. INTERNATIONAL MUON COLLIDER collaboration, *Interim report for the International Muon Collider Collaboration (IMCC)*, [2407.12450](#).
8. CEPC STUDY GROUP collaboration, *CEPC Technical Design Report: Accelerator*, *Radiat. Detect. Technol. Methods* **8** (2024) 1 [[2312.14363](#)].
9. J. Reuter, T. Han, W. Kilian, N. Kreher, Y. Ma, T. Striegl et al., *Precision test of the muon-Higgs coupling at a high-energy muon collider*, *PoS ICHEP2022* (2022) 1239 [[2212.01323](#)].
10. T. Han, A. K. Leibovich, Y. Ma and X.-Z. Tan, *Higgs decay to charmonia and the charm-quark Yukawa coupling*, *PoS ICHEP2022* (2022) 517 [[2211.10727](#)].
11. K. M. Black et al., *Muon Collider Forum report*, *JINST* **19** (2024) T02015 [[2209.01318](#)].
12. T. Han, Y. Ma and K. Xie, *Electroweak fragmentation at high energies: A Snowmass White Paper*, in *Snowmass 2021*, 3, 2022, [2203.11129](#).
13. J. M. Campbell et al., *Event Generators for High-Energy Physics Experiments*, *SciPost Phys.* **16** (2024) 130 [[2203.11110](#)].
14. MUON COLLIDER collaboration, *A Muon Collider Facility for Physics Discovery*, [2203.08033](#).
15. C. Aime et al., *Muon Collider Physics Summary*, [2203.07256](#).
16. MUON COLLIDER collaboration, *The physics case of a 3 TeV muon collider stage*, [2203.07261](#).
17. ILC INTERNATIONAL DEVELOPMENT TEAM collaboration, *The International Linear Collider: Report to Snowmass 2021*, [2203.07622](#).

SEMINARS AND COLLOQUIUM

- | | |
|---|---------------|
| 1. <i>A New Frontier: Charting the Course for a 10 TeV Muon Collider</i>
SOTU Seminar, Tata Institute of Fundamental Research, Mumbai, India | October 2025 |
| 2. <i>Precision physics at the future muon colliders</i>
Theory Colloquium, DESY, Germany | May 2025 |
| 3. <i>Probe the Yukawa interactions of the 2nd generation fermions at high-energy colliders</i>
TDLI/INPAC Joint Theory Seminar, Shanghai Jiao Tong University | July 2024 |
| 4. <i>Physics opportunities and challenges at future multi-TeV lepton colliders</i>
HEP Seminar, Nankai University | December 2023 |

5. <i>Physics opportunities and challenges at future multi-TeV lepton colliders</i> HEP Seminar, Chongqing University	December 2023
6. <i>Physics opportunities and challenges at future multi-TeV lepton colliders</i> HEP Seminar, Central South University	December 2023
7. <i>Physics opportunities and challenges at future multi-TeV lepton colliders</i> HEP Seminar, Hunan University	December 2023
8. <i>Physics opportunities and challenges at future multi-TeV lepton colliders</i> HEP Seminar, Shandong University	December 2023
9. <i>Physics opportunities and challenges at future multi-TeV lepton colliders</i> HEP Seminar, University of Science and Technology of China (USTC)	December 2023
10. <i>Physics opportunities and challenges at future multi-TeV lepton colliders</i> HEP Seminar, Nanjing Normal University	December 2023
11. <i>Physics opportunities and challenges at future multi-TeV lepton colliders</i> HEP Seminar, Southeast University	December 2023
12. <i>Physics opportunities and challenges at future multi-TeV lepton colliders</i> HEP Seminar, Fudan University	December 2023
13. <i>Physics opportunities and challenges at future multi-TeV lepton colliders</i> HEP Seminar, Nanjing University	December 2023
14. <i>Physics opportunities and challenges at future multi-TeV lepton colliders</i> HEP Seminar, Tsung-Dao Lee Institute (TDLI), Shanghai Jiao Tong University	December 2023
15. <i>Physics opportunities and challenges at future multi-TeV lepton colliders</i> HEP Seminar, Tsinghua University	November 2023
16. <i>Physics opportunities and challenges at future multi-TeV lepton colliders</i> HEP Seminar, Peking University	November 2023
17. <i>Physics opportunities and challenges at future multi-TeV lepton colliders</i> HEP Seminar, Institute of Theoretical Physics, Chinese Academy of Sciences	November 2023
18. <i>Physics opportunities and challenges at future multi-TeV lepton colliders</i> HEP Theory Seminar, IHEP, Chinese Academy of Sciences	November 2023
19. <i>Bread and butter physics at future multi-TeV lepton colliders</i> NHETC Theory Seminar, Rutgers University	May 2023
20. <i>Measuring the Yukawa couplings: Towards the 2nd generation fermions</i> HEP Lunch Seminar, University of Chicago	May 2023
21. <i>Electroweak Tevatron: High-Energy lepton colliders</i> Theoretical Physics Seminar, Fermilab	May 2023
22. <i>Bread and butter physics at future multi-TeV lepton colliders</i> LEPP Theory Seminar, Cornell University	May 2023
23. <i>Measuring the Yukawa couplings: Towards the 2nd generation fermions</i> HEP Theory Seminar, University at Buffalo	May 2023
24. <i>Determine the Yukawa couplings of the second generation fermions at high-energy colliders</i> (Remote) HEP Theory Seminar, Argonne National Laboratory (ANL)	January 2023

- | | |
|---|----------------|
| 25. <i>Determine the Yukawa couplings of the second generation fermions at high-energy colliders</i>
(Remote) TDLI/INPAC Joint Theory Seminar, Shanghai Jiao Tong University | December 2022 |
| 26. <i>Determine the Yukawa couplings of the second generation fermions at high-energy colliders</i>
(Remote) Particle Physics Seminar, Chongqing University | December 2022 |
| 27. <i>Determine the Yukawa couplings of the second generation fermions at high-energy colliders</i>
(Remote) Theoretical Physics Seminar, Shandong University | December 2022 |
| 28. <i>Phenomenology at high-energy colliders</i>
Bologna HEP Theory Journal Club, INFN Bologna & University of Bologna | November 2022 |
| 29. <i>The partonic picture and the SM expectation of high-energy lepton colliders</i>
HEP Seminar, University of Notre Dame | September 2022 |
| 30. <i>The partonic picture and the SM expectation of high-energy lepton colliders</i>
HEP Seminar, University of Wisconsin-Madison | September 2022 |
| 31. <i>The partonic picture and the SM expectation of high-energy lepton colliders</i>
HEP Special Seminar, University of Michigan | September 2022 |
| 32. <i>The partonic picture and the SM expectation of high-energy lepton colliders</i>
HEP Seminar, Northwestern University | September 2022 |
| 33. <i>Higgs decay to charmonia and the charm-quark Yukawa coupling</i>
HEP Seminar, Michigan State University | September 2022 |
| 34. <i>Higgs decay to charmonia and the charm-quark Yukawa coupling</i>
HEP Seminar, Washington University in St. Louis | September 2022 |
| 35. <i>The partonic picture and the SM expectation of high-energy lepton colliders</i>
HEP Seminar, University of Minnesota | August 2022 |
| 36. <i>Higgs decay to J/ψ via c-quark fragmentation</i>
(Remote) Nuclear Physics Seminar, UCLA | May 2022 |
| 37. <i>Higgs decay to charmonia and the charm quark Yukawa</i>
PITT PACC Group Seminar, University of Pittsburgh | March 2022 |
| 38. <i>Multi-boson production and the muon Yukawa coupling</i>
(Remote) HEP Journal Club, University of Utah | October 2021 |
| 39. <i>Multi-boson production and the muon Yukawa coupling</i>
PITT PACC Group Seminar, University of Pittsburgh | September 2021 |
| 40. <i>Parton contents of a lepton at high energies</i>
(Remote) Particle Theory Seminar, Carleton University | May 2021 |
| 41. <i>The partonic picture at high-energy lepton colliders</i>
(Remote) SLAC EPP Theory Seminar, SLAC | April 2021 |
| 42. <i>The partonic picture at high-energy lepton colliders</i>
(Remote) Particle Theory Seminar, Shandong University | April 2021 |
| 43. <i>Parton contents of a lepton at high energies</i>
(Remote) HEP Seminar, Oklahoma State University | April 2021 |

- | | |
|---|----------------|
| 44. <i>QCD jet production at high energy lepton colliders</i>
(Remote) PITT PACC Group Seminar, University of Pittsburgh | March 2021 |
| 45. <i>High energy lepton collisions and electroweak PDFs</i>
(Remote) Particle Theory Seminar, Carleton University | October 2020 |
| 46. <i>High energy lepton collisions and electroweak PDFs</i>
(Remote) PITT PACC Group Seminar, University of Pittsburgh | September 2020 |
| 47. <i>How much do we need polarized PDFs?</i>
PITT PACC Group Seminar, University of Pittsburgh | October 2019 |
| 48. <i>Renormalization scheme uncertainties in high order perturbative QCD results</i>
PITT PACC Group Seminar, University of Pittsburgh | March 2019 |

CONFERENCE AND WORKSHOP TALKS

- | | |
|--|---------------|
| 1. <i>Beyond the Standard Model at Future Colliders</i>
(Invited) Parallel talk at LHCP 2026, Sorbonne Université, France | May 2026 |
| 2. <i>Chasing 2HDM via electroweak corrections at e^+e^- colliders</i>
Parallel talk at Pheno 2026, University of Pittsburgh | May 2026 |
| 3. <i>Toward precision Higgs at FCC-ee and jet physics at the muon collider</i>
MADGRAPH5_aMC@NLO meeting 2026, Heidelberg, Germany | February 2026 |
| 4. <i>Precision physics at the future muon collider</i>
(Invited) Parallel talk at the 2nd Annual US Muon Collider Meeting, University of Chicago | August 2025 |
| 5. <i>Higgs-muon interactions at a multi-TeV muon collider</i>
GGI workshop: <i>Exploring the energy frontier with muon beams</i> , Florence, Italy | July 2025 |
| 6. <i>Light Axion-Like Particles at Future Lepton Colliders</i>
Parallel talk at Pheno 2025, University of Pittsburgh | May 2025 |
| 7. <i>Probing electroweak NLO corrections and Sudakov logarithms at multi-TeV muon colliders</i>
Parallel talk at IMCC and MuCol Annual Meeting 2025, DESY, Germany | May 2025 |
| 8. <i>Precision physics at the future muon collider</i>
MADGRAPH5_aMC@NLO meeting 2025, CERN | February 2025 |
| 9. <i>Higgs-muon interactions at a multi-TeV muon collider</i>
Parallel talk at Higgs 2024, Uppsala University, Sweden | November 2024 |
| 10. <i>Charm and bottom Yukawa couplings via quarkonia production at HL-LHC</i>
Parallel talk at Higgs 2024, Uppsala University, Sweden | November 2024 |
| 11. <i>Electroweak radiation picture of the future multi-TeV muon collider</i>
Early Career Researchers & Muon Colliders (Online Event) | August 2024 |
| 12. <i>Higgs-muon interactions at a multi-TeV muon collider</i>
Parallel talk at ICHEP 2024, Prague, Czech Republic | July 2024 |
| 13. <i>Towards a Muon Collider: III. Higgs Physics</i>
Frontier Physics Working Month, South China Normal University | June 2024 |
| 14. <i>Towards a Muon Collider: II. The Partonic Picture</i>
Frontier Physics Working Month, Shandong University (Qingdao) | June 2024 |

15. *Towards a Muon Collider: I. The General Picture* June 2024
Frontier Physics Working Month, Peking University
16. *Probing Higgs-Muon Interactions at Multi-TeV Collider* March 2024
Parallel talk at IMCC and MuCol Annual Meeting 2024, CERN
17. *Multiple boson production at high-energy muon colliders to probe the Higgs-muon coupling* December 2023
Parallel talk at Higgs 2023, IHEP, Beijing
18. *Higgs decay to quarkonia and the Yukawa couplings* November 2023
Parallel talk at Higgs 2023, IHEP, Beijing
19. *Muon colliders and Weak PDFs* September 2023
MADGRAPH5_aMC@NLO meeting 2023, Gargnano, Lake Garda, Italy
20. *Muon Yukawa couplings at the high-energy muon collider* May 2023
Parallel talk at Pheno 2023, University of Pittsburgh
21. *Electroweak LHC: High-energy lepton colliders* April 2023
Invited talk at PIKIMO Spring 2023, Ohio State University
22. *EW and QCD physics at the muon collider* December 2022
Parallel talk at Milan Christmas Meeting 2022, Milan, Italy
23. *Higgs decay to charmonia and the charm-quark Yukawa coupling* November 2022
Parallel talk at the Higgs 2022 Conference, Pisa, Italy
24. *EW and QCD physics at the muon collider* October 2022
Parallel talk at Muon Collider Collaboration Meeting 2022, CERN
25. *Higgs decay to charmonia via c-quark fragmentation* September 2022
Invited plenary talk at QWG 2022, GSI Darmstadt, Germany
26. *Higgs decay to charmonia and the charm-quark Yukawa coupling* September 2022
(Remote) Invited talk at the SYSU-PKU Collider Physics forum For Young Scientists
27. *Higgs decay to J/ψ via c-quark fragmentation* July 2022
(Remote) Parallel talk at ICHEP 2022, Bologna, Italy
28. *Higgs decay to J/ψ via c-quark fragmentation* May 2022
Parallel talk at Pheno 2022, University of Pittsburgh
29. *Multi-boson production and the muon Yukawa coupling* April 2022
Contributed talk at APS April Meeting 2022, New York
30. *Multi-boson production and the muon Yukawa coupling* December 2021
PIKIMO 11, University of Pittsburgh
31. *Electroweak parton distributions and fragmentations for high-energy lepton colliders* October 2021
(Remote) Snowmass EF04 Topical Group Community Meeting
32. *Higgs boson decay to J/ψ via c-quark fragmentation* October 2021
(Remote) Parallel talk at Higgs 2021 Conference, Stony Brook University
33. *The partonic picture at high-energy lepton colliders* August 2021
(Remote) Parallel talk at SUSY 2021, Shanghai
34. *QCD jet production at a high energy muon collider* July 2021
(Remote) Parallel talk at EPS-HEP 2021, DESY

- | | |
|---|----------------|
| 35. <i>Quark and gluon contents of a lepton at high energies</i>
(Remote) Parallel talk at the DPF meeting, Florida State University | July 2021 |
| 36. <i>Quark and gluon contents of a lepton at high energies</i>
(Remote) Parallel talk at Pheno 2021, University of Pittsburgh | May 2021 |
| 37. <i>The partonic picture at high-energy lepton colliders</i>
(Remote) Parallel talk at PPC 2021, University of Oklahoma | May 2021 |
| 38. <i>Electroweak parton distribution functions at a high-energy muon collider</i>
(Remote) Contributed talk at APS April Meeting 2021 | April 2021 |
| 39. <i>QCD jet production at a high energy muon collider</i>
(Remote) Talk at Muon Collider Physics and Simulation Meeting, CERN | March 2021 |
| 40. <i>The electroweak parton distribution functions - Necessity and application</i>
(Remote) Student talk at Theoretical Advanced Study Institute (TASI 2020) | June 2020 |
| 41. <i>The electroweak parton distribution functions</i>
(Remote) Parallel talk at Pheno 2020, University of Pittsburgh | May 2020 |
| 42. <i>QCD Scale-setting problem in Future Chinese Collider physics</i>
Parallel talk at CEPC-SppC Study Group Meeting, IHEP, Beijing | September 2015 |

CONFERENCES AND WORKSHOPS ATTENDED

-
- | | |
|--|----------------|
| 1. The 14th Annual Large Hadron Collider Physics conference (LHCP 2026)
Sorbonne Université, France | May 2026 |
| 2. Phenomenology Symposium 2025 (Pheno 2026), University of Pittsburgh | May 2026 |
| 3. MADGRAPH5_aMC@NLO meeting 2026, Heidelberg, Germany | February 2026 |
| 4. Second Annual US Muon Collider Meeting, University of Chicago | July 2025 |
| 5. GGI workshop: <i>Exploring the energy frontier with muon beams</i> , Florence, Italy | July 2025 |
| 6. Phenomenology Symposium 2025 (Pheno 2025), University of Pittsburgh | May 2025 |
| 7. IMCC and MuCol Annual Meeting 2025, DESY, Germany | May 2025 |
| 8. MADGRAPH5_aMC@NLO meeting 2025, CERN | February 2025 |
| 9. Higgs 2024, Uppsala University, Sweden | November 2024 |
| 10. Early Career Researchers & Muon Colliders (Online Event) | August 2024 |
| 11. The 42nd International Conference on High Energy Physics (ICHEP 2024)
Prague, Czech Republic | July 2024 |
| 12. Frontier Physics Working Month 2024, Peking University | June 2024 |
| 13. IMCC and MuCol Annual Meeting 2024, CERN | March 2024 |
| 14. Higgs 2023, IHEP, Beijing | November 2023 |
| 15. MADGRAPH5_aMC@NLO meeting 2023, Gargnano, Lake Garda, Italy | September 2023 |
| 16. Muon Collider Collaboration Meeting 2023, IJCLab in Orsay, France | May 2023 |
| 17. Phenomenology Symposium 2023 (Pheno 2023), University of Pittsburgh | May 2023 |
| 18. PIKIMO Spring 2023, Ohio State University | April 2023 |

19. Milan Christmas Meeting 2022, Milan, Italy	December 2022
20. Higgs 2022, Pisa, Italy	November 2022
21. Muon Collider Collaboration Meeting 2022, CERN	October 2022
22. The 15th International Workshop on Heavy Quarkonium (QWG 2022) GSI Darmstadt, Germany	September 2022
23. SYSU-PKU Collider Physics forum For Young Scientists (remote)	September 2022
24. Snowmass Community Summer Study Workshop (Snowmass 2022), Seattle	July 2022
25. XLI International Conference on High Energy Physics (ICHEP 2022), Bologna, Italy	July 2022
26. LoopFest XX, University of Pittsburgh	May 2022
27. Phenomenology Symposium 2022 (Pheno 2022), University of Pittsburgh	May 2022
28. APS April Meeting 2022, New York	April 2022
29. PIKIMO 11, University of Pittsburgh (hybrid)	December 2021
30. Higgs 2021 Stony Brook University & Brookhaven National Laboratory (remote)	October 2021
31. The XXVIII International Conference on Supersymmetry and Unification of Fundamental Interactions (SUSY 2021), Shanghai (remote)	August 2021
32. European Physical Society Conference on High Energy Physics 2021 (EPS-HEP 2021) DESY (remote)	July 2021
33. 2021 Meeting of the Division of Particles and Fields of the APS (DPF21) Florida State University (remote)	July 2021
34. Phenomenology Symposium 2021 (Pheno 2021) University of Pittsburgh (remote)	May 2021
35. XIV International Workshop on Interconnections between Particle Physics and Cosmology (PPC 2021), University of Oklahoma (remote)	May 2021
36. APS April Meeting 2021 (remote)	April 2021
37. Muon Collider Physics and Simulation Meeting (remote)	March 2021
38. PITT PACC Workshop: Muon collider physics University of Pittsburgh (remote)	November 2020
39. Phenomenology Symposium 2020 (Pheno 2020) University of Pittsburgh (remote)	May 2020
40. Phenomenology Symposium 2020 (Pheno 2019), University of Pittsburgh	May 2019
41. PITT PACC Workshop: BSM circa 2020, University of Pittsburgh	March 2019
42. Phenomenology Symposium 2020 (Pheno 2018), University of Pittsburgh	May 2018
43. Phenomenology Symposium 2020 (Pheno 2017), University of Pittsburgh	May 2017
44. The CEPC-SppC Study Group Meeting Institute of High Energy Physics (IHEP), Beijing	September 2015

SUMMER SCHOOLS ATTENDED

1. **CTEQ 2022**, University of Pittsburgh July 2022
CTEQ School on QCD and Electroweak Phenomenology
2. **SSI 2021**, SLAC August 2021
49th SLAC SUMMER INSTITUTE: The Higgs State Fair
3. **HCPSS 2020**, Fermilab August 2020
15th annual Fermilab-CERN Hadron Collider Physics Summer School
4. **TASI 2020**, University of Colorado Boulder June 2020
The Obscure Universe: Neutrinos and Other Dark Matters
5. **CTEQ 2019**, University of Pittsburgh July 2019
CTEQ School on QCD and Electroweak Phenomenology
6. **CTEQ 2017**, University of Pittsburgh July 2017
CTEQ School on QCD and Electroweak Phenomenology